

Observatories as Transparency Instruments for Projects

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Abstract – The observation, in the context of the organization and the projects, has been defined, in the recent literature, by the transparency construct, and the development of tools to systematize the transparency has presented as a challenge to the organizations. In this sense, the observatories are presented as information systems that support the collection, organization, storage, analysis, and observation publications, providing transparency. In this scenario, this research had the purpose of discuss the observatories construction as transparency mechanism to projects. To this, a set of three pilot projects of projects observatories were developed. This research is classified as an exploratory research with a qualitative approach, performed from participant observation and archival research. The construction these observatories made possible to know and test the concept of projects observatories in practice, in addition to identifying its characteristics and variability. It was also identified an initial set of functions and components of the observatories, in addition of watching the benefits and difficulties of its construction.

Keywords - Observation, Transparency, Observatories, Projects.

I. INTRODUCTION

Observation has always been a fundamental element of management and, in fact, of the everyday life. With time, many theorists used different constructs to capture their interests for the observation. The most recent construct in the attempt of defining the observation in the context of management is the transparency [1]. In the projects context, the observation translated by the transparency phenomena, has gain power in the last years. According to [2], as there are not business in the world without projects, there cannot be success in the projects without transparency.

Transparency can provide a lot of advantages to the organizations, among them, improvement of the communication, sharing knowledge and learning of the organizations and its employees, in addition to improve the efficiency of the market and consolidate relationships between organizations [3]. The development of mechanisms to systematize transparency is a challenge to the organizations [4]. According to [5], studies must be developed with the goal to investigate how to increase and

strengthen ties between organizations and their external environments, offering better services and establishing an effective dialog between them using the technology, specially the information systems.

In this sense, observatories present themselves as information systems that support collection, organization, storage, analysis and observation publication, providing transparency [6,7]. In literature, it is possible to find a diversity of published papers about observatories, applied to the most variable themes and areas of knowledge [8-10]. In this scenario, this research had the purpose of discuss the observatories development as transparency mechanism to projects. To this, a literature review on observatories was made and, after that, a set of pilot projects about projects observatories was developed. The construction of these pilot projects has made possible to know and test the concept of projects observatories in practice.

In [11] studies related to the development of projects observatories were presented, in this work are presented initial guidelines for the development of a projects observatory, the Universal Project Observatory. The studies by [11] served as a starting point for the execution of this research. In this sense, in this research we seek to evolve the understanding about the projects observatories presented by [11], based on the development of new pilot projects of observatories. In addition to this Section, this paper is organized into other four Sections. In Section 2, the main concepts related to thematic are presented. Section 3 describes the methodology used to execute this research. The results obtained from execution of studies are presented in Section 5. Finally, Section 6 presents the concluding remarks.

II. LITERATURE REVIEW

A. Observation in the context of projects and management

Demand for observation in business came in the ancient civilizations, with the accounting systems. However, the goal of the observation in business was not just to learn, like the natural sciences, but also to influence and, finally, control. Systematic observation of learning and control

came to the forefront of management practice when Frederick W. Taylor sought to apply the scientific method and, therefore, systematic observation, to organizational problems [1]. Thereby, we can understand that observation has been used as a management support instrument since before the beginnings of administration as a science.

Nowadays, organizations live a new era of the management applied of the observation - transparency. Transparency logic is based fundamentally on the premise that a more precise perception of the others improves learning and control and, therefore, improve performance [1]. Observation, understood by the construct of transparency, also stands out as a fundamental element to the projects management. Transparency is a universal characteristic of the development and implantation processes of projects [2].

The transparency can improve the view of organizational processes and information, offering knowledge information about them, reducing the possibility for information omission, allowing the control over products and services, making the research easier and increasing trust between organizations and society [4]. Therefore, in the private and public contexts, transparency establishes a set of aspects that suggest the existence of policies, procedures and technologies to provide not only access, but also use, quality and understanding of processes and information [12].

B. Observatories as observation instruments

There is no consensus definition in the literature for the concept of “observatories”, although some studies relate them to observation instruments [9], collection [13], consolidation [9], storage [14], analysis [15], sharing [15] and disclosure [10] of data, information and knowledge about a sector or area of knowledge. Sakata et al. [7] adds that these observatories can be seen as instruments that promote transparency.

When the term observatory is mentioned, what comes to people's mind are the observatories connected to natural phenomena. However, the concept of observatory connected to nature was initially transported to a social area, and with this, a greater coverage [16]. The business area also started to benefit with the development of the observatories, once they are responsible for the collection and dissemination of strategic information to assist in decision-making processes [16]. As can be seen, the term has undergone a recent expansion of its meaning or of its semantic field since its original attribution to the observatories of natural phenomena.

According to [17], observatories can be understood as socio-technical systems, covering a set of users and technologies in a social context. [18] complement that an observatory can be of three types of nature: (i) organizational unit - as an element of the organization (department, core or center) that performs the function of an observatory; (ii) mechanism or process - is a device

where the functions of an observatory are performed by a responsible group; (iii) instrument - in the case of the technology or tool used to fulfill the function of an observatory. It is possible to identify in the literature a set of benefits that observatories can provide for a particular organization, sector or area of knowledge, among these benefits are:

- Access to data, information and knowledge available on a given subject [18].
- Facilitate communication, cooperation and collaboration between people [19].
- Contribute to improving management and planning, optimizing and supporting decision making [9].
- Support society and, in this context, they can improve democracy, increase transparency, inhibit corruption and fraud, support the design of public policies, mobilize citizens, promote accountability, and promote social participation and control [19].

III. METODOLOGY

We use the taxonomy of [20] to characterize this research. Considering its results, this paper is classified as applied research. Regarding the research objective, this paper can be considered an exploratory research. About the research process, it can be considered a qualitative research. Lastly, data was collected from the literature revision on the observatories, the reports produced by the teams that participated in the studies (archive researches) and the observation process performed by the authors of this paper during studies execution (participant observation).

This research was developed in 2 steps, as presented in Figure 1. Step 1, a literature revision about the observatories was made. The revision had the goal of understanding how the observatories are characterized in the literature. The result of this first step served as input for the execution of the following step. Step 2 consisted on the execution of 3 studies, which had the goal of development of the pilot projects of projects observatories. We performed these studies in one subject of the post-graduation program in Computer Science at the Computer Center of the Federal University of Pernambuco.

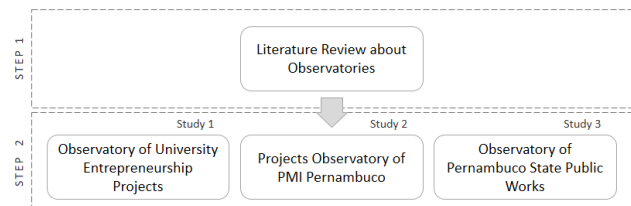


FIGURE 1. RESEARCH STEPS

We executed Step 2 in a subject dedicated to the development of observatories, having as main theme the “Development of Projects Observatories”. In this subject,

divided into three groups, the students developed pilot projects of the following observatories: Observatory of University Entrepreneurship Projects (Study 1), Projects Observatory of PMI Pernambuco (Study 2) and Observatory of Pernambuco State Public Works (Study 3). The results of step one (literature revision on observatories) e the results of studies about the development of projects observatories presented in [11] were used as inputs to the construction of these pilot projects.

Studies developed in the third step, were executed in the second semester of 2019, and 9 students participated. All the participants were about 38 years old, all male, and were master or doctoral students. The participants had computer (57%) and administration (43%) as degrees, and about 16 years of experience in the areas. Most of the participants (71%) informed that they hold or have held some management position, among them, project management and general manager of an organization. All of them informed having previously participated in a project in the areas of information and communication technology, products development, education, events and research and development.

IV. RESULTS

A. Study 1: Observatory of University Entrepreneurship Projects

The team participating in Study 1 proposed the Observatory of University Entrepreneurship Projects (OPEU, - acronym in Observatório de Projetos de Empreendedorismo Universitário) as a place for observation, analysis and reflection of the movements in entrepreneurship and entrepreneurial initiatives in university centers, promoting the collaboration between teachers, students, researchers and entrepreneurs to build content and knowledge about the theme, driven by inspiring cases.

On its initial version, this observatory included university entrepreneurship projects developed on the subject of "Development Project" of the graduation in Computer Science at the Center for Informatics of the Federal University of Pernambuco. This subject offers students the opportunity to develop a computer system, encouraging the emergence of entrepreneurial initiatives.

These entrepreneurial initiatives developed at the university level can serve as a reference and stimulus for the emergence of new entrepreneurial ideas from the students. However, the knowledge and experience of these entrepreneurial projects, in case they are not put somewhere, could be lost or limited just to the participants of the initiatives, that many times turn the project into a startup or yet a new company. Therefore, the OPEU propose to be this environment to retain this knowledge, publicizing the projects developed and encouraging

collaboration between those involved. The observatory was available to access through a web page¹.

In this observatory, any previously registered user can publish information about a project. For that, a project registration form was created, where the user can enter the following information: name, type, institution, description, start and end date, status, website, repository, partners, main sector benefited, motivation, objective, budget, stakeholders, among others. Once published, the project can be viewed by any user of the observatory (without prior registration). Besides that, users can also connect via Facebook, to interact with the observatory through comments in the pages of the projects.

Beyond the possibility of register and see the information on the projects, this observatory has areas to the dissemination of news and events related to university entrepreneurship. Finally, an area reserved for publication of studies and research developed on the theme was also created. Before the launching of the observatory, the team made an initial load with information on 98 projects developed on the subject "Development Projects", between the years 2007 and 2019.

During the observatory development, the team reported difficulties related to the data collection, especially with the historical data of the projects developed in the previous years. A large amount of important information on the projects had been lost, or were scattered in the most variable sources and formats. In addition, the heterogeneity of the data made it difficult to structure a minimum set of project metadata.

B. Study 2: Projects Observatory of PMI Pernambuco

The team participating in Study 2 proposed the Projects Observatory of PMI Pernambuco as a solution to provide visibility to the projects and actions carried out by the PMI chapter of the state of Pernambuco (PE) to all society, as well as to generate interaction between its members and internal chapter directors. Currently, much of the information produced during the planning and execution of the projects and actions developed by PMI-PE are centralized in the chapter's directorates. However, giving visibility to this information can generate good results, both for the more than 300 members currently affiliated, as well as for the chapter management itself.

The observatory developed for the PMI-PE is an online platform based on a software to provide its operation and ensure interaction with its respective users. The goal of the team was to produce a web tool for: publication and sharing of the projects; enable access to chapter affiliates (restricted area); enable access to society (public area); allow interaction among affiliates with the chapter projects; enable documents storing, as repository of information and artifacts; and allow to view some metrics in relation to the

¹ Available in: <https://opeu.com.br/>

projects. The observatory was available to access through a web page².

Unlike the other studies presented here, in this observatory only a smaller community, composed of members affiliated with PMI-PE, has access to all data on the projects developed. Whereas, users not affiliated with the chapter have access to a limited amount of information about these projects. Therefore, in order to make this possible, the team planned to build an exclusive area for access to PMI-PE affiliated members.

To non-affiliated users, the observatory shows high-level information on the projects, like name and amount of projects currently performed by the chapter. In addition to that, a set of templates used to document the projects developed in the Chapter is also available to this type of user. To the PMI-PE users a restricted area was built. Through a prior registration, this user can publish information on projects developed by the PMI-PE, it is even possible to attach medias and documents to those projects. In addition, these users will be able to comment on projects published in the observatory and view statistics on those projects.

The team carried out an initial load on the tool with information on 50 projects for the years 2016 to 2019. This observatory was designed to store the following attributes of the projects: name, description, place, start and end date, participants, budget, cost, results, project manager, status, among others. The team reported some issues while building the observatory, about the data confidentiality specially. Not all information on the projects developed by the PMI chapter could be made available to the public; thus, considerable effort was required to identify what information could or could not be shared at the observatory.

C. Study 3: Observatory of Pernambuco State Public Works (ObrasPE)

The team participating in Study 3 proposed the Observatory of Pernambuco State Public Works as an open platform that has all data available about the works, with many visualization resources. In addition, this platform must provide a space for interaction between users and those users with the data and visualizations produced.

This observatory presents itself as a relevant instrument to the public administration point of view, through which it will now have a tool that allows it to comply with the principles of transparency and advertising. In addition to being relevant to organized society and, ultimately, the citizen, who will have an effective instrument in the exercise of citizenship. Once the observatory was built, an initial load with 1.512 work information was made. This load was made through information extracted from a database available for download by TCE-PE in the website.

The team defined a set of minimum attributes for each work, such as title, place of execution, start date, status, category, term, added term and deadline.

A user will be able to register to interact with the observatory, through comments on the details pages of the work. In addition to writing in the comments, something related to the work, the user could also attach photos to that comment if they want to. It is also available to the users (registered or not), access to a set of visualizations on the collected data, making possible to make since simple filters by keywords, categories, municipality or managing units, to more sophisticated visualizations, like a heat map by amount of paralyzed works in the cities. In addition, the observatory present a set of graphs with general information from the state and with particular information from each municipality, related to: total amount of paralyzed works, paralyzed works by category, management units and executing companies with the greatest number of works paralyzed, among others. The observatory was available to access through a web page³.

While constructing the observatory, the team reported some difficulties related to the data collection. Initially, this observatory had been planned to contemplate information on all the works (paralyzed or not) that would have been executed in the state of Pernambuco. However, it was not found a public database with that information. There was an attempt to request the data from the government; however, due to the limited time for the construction of the observatory and the difficulty of obtaining feedback from the requests, the scope of the project had to be reduced. In addition to the lack of transparency, the team also faced problems related to the low quality of the data, especially in the data related to the geolocation of the works.

D. DISCUSSION OF RESULTS

The results found, from the studies made, allow us to identify an initial concept to the projects observatories. In this perspective, the observatories developed here were presented as an instrument or mechanism of transparency, based on a computational system, that allow a community to: observe, analyze and think about the projects; collect and produce big amount of data on the projects; discuss and show thoughts on the projects; follow projects; gather and manage various information about projects; and share knowledge about the projects. Studies also show that the concept can be applied in the context of public projects, and private projects. This concept is broadly in line with the definitions of observatories identified in the literature [9,10,13,14,15].

From the studies, it is also possible to identify a set of reflections and lessons learned related to their characteristics, variability, functions, components and

² Available in: <http://pmi-op.herokuapp.com/>

³ Available in: <https://bitly.com/ISNse>

construction difficulties. The first point of reflection is related to the scope of these observatories. The developed observatories have a more limited scope, contemplating projects developed by an organization, projects belonging to a geographical region or, yet, projects related to a specific theme. In this sense, the scope of a projects observatory must be defined at the time of its conception. We understand that the bigger is the scope; bigger will be the challenge for its construction.

In all of the studies, were planned and developed mechanisms of repository to storage project data (such as, description, nature, size, place of execution, budget, planned and developed activities, stakeholders, status, risks, schedule, life cycle, results, etc.), and the data about observations that can be done by the users in the registered projects (text, photo, video, etc.). Thus, the development of projects observatories presupposes the construction of projects repositories, which must be prepared to store structured and unstructured data. The need to build data repositories to support the observatories is presented in the literature by [21]. In addition, considering that not all project information may be contained in the observatories, the repositories must also be prepared to store partial information about the projects.

It is possible to identify some variations in the characteristics of the developed observatories. The first one is related to the opening level of the observatory for its users. In the studies two types of observatories were developed: open and semi-open. In the open observatories, there is not restriction to the participation of the users, meaning, they have complete access to all functionalities of the observatory. In the semi-open observatories, some functionalities are exclusive to a certain group of users. However, in all the studies it was possible to notice that the projects observatories depends on the existence of a community of users, beyond the development team of the projects.

The studies also pointed out variations in the way users interact with observatories. It was possible to identify three types of interactions: (1) users can register data on the projects; (2) users can make observations on the projects (write comments and attach photos or videos); and (3) users can see information and analysis produced on the data collected. Each developed observatory has implemented one or more of these forms of interaction.

It is possible to notice that the way the observatories implemented data collection also varied. Some observatories built instruments to make the collection manually, seeking project data from their managers and allowing them to register this information in the observatory. Others planned or built instruments to collect this information in an automated way, by building data collectors on the web or by implementing mechanisms that consume data via services offered by other tools.

Studies have also allowed us to identify an initial set of functions of the projects observatories, confirmed by the observatories literature [19], related to: information collection and dissemination; treatment and validation of information; construction and dissemination of analyzes; and coordination of user relationships. Besides that, it was also possible to identify some components of these observatories, such as collection instruments and data validation, analysis construction mechanisms, and dissemination tools.

Finally, studies also showed the following difficulties related to the observatories construction, in special to the process of data collection: absence or low amount of project data; information about projects stored in a decentralized way in the most diverse sources, bureaucracy for data release; data confidentiality; lack of transparency, especially in the public context, heterogeneity of data; and the existence of data stored in non-digital formats.

The table 1 summary of set of learned lessons about the projects observatories and its constructions process.

TABLE I. LEARNED LESSONS WITH THE EXECUTION OF STUDIES

Learned Lessons	Studies
The more comprehensive the scope of the projects observatory, the greater the challenge of its construction. However, also we can benefits be achieved with the development of smaller scope observatories.	1, 2, and 3
The development of projects observatories requires the construction of repository mechanisms to store project data.	1, 2, and 3
The repositories must store structured and unstructured data about the projects.	1, 2, and 3
The repositories must store partial information about the projects.	1, 2, and 3
Projects observatories depend on the existence of a user community, which goes beyond the project development team.	1, 2, and 3
The projects observatories can be open, where there is not restriction to the participation of the users, meaning, they have complete access to all functionalities of the observatory.	1 and 3
The observatories can be semi-open, where its functionalities are exclusive access to a certain group of users.	2
Users can interact with projects observatories by registering project data, making observations about the projects (write comments and attach photos or videos) and/or viewing information and analysis produced on the collected data.	1, 2, and 3
Projects observatories can collect data manually or automatically.	1, 2, and 3
The projects observatories can present the following function set: information collection and dissemination, treatment and validation of information, construction and dissemination of analyzes, and coordination of user relationships.	1, 2, and 3
The projects observatories have components related to data collection and validation, storage, building analyzes, and disclosure.	1, 2, and 3
Data collection is a critical factor that can cause difficulties limiting the development of projects observatories.	1, 2, and 3

V. FINAL CONSIDERATIONS

The observatories present themselves as instruments of transparency that allow the collection, consolidation,

storing, analysis, sharing and data disclosure, information and knowledge about a sector or area of knowledge. This way, the research had the goal to propose some reflections about observatory construction applied to the projects context.

To achieve the objective of this work, we present three studies with the purpose of building pilot projects for projects observatories. The results here presented were extracted from the reports made by the participants of the studies and observations made by the authors of this work. The studies presented the following pilot projects of projects observatories: Observatory of University Entrepreneurship Projects, Projects Observatory of PMI Pernambuco, and Observatory of Pernambuco State Public Works.

These studies did not have as prime objective the construction and availability of the projects observatories to its users, despite that, we achieved some results were in this direction. However, the pilot projects developed in this paper can serve as a basis for the development of new versions for these observatories. From the execution of these studies, it was possible to identify a set of learned lessons (summarized in Table 1) about the projects observatories and its constructions process. The results obtained from the execution of this work can assist in the development of new project observatories. And, it is hoped that these tools can be used as a resource to promote transparency in the context of projects.

The conclusions presented here bring us new research challenges. An assessment based on the exhaustive use of the built observatories may bring us new results. In addition, we can made new reflections from the development of new projects observatories. Researches about information technologies that support collection, storage and data dissemination, information and knowledge about the projects may also be developed. Moreover, finally, studies that lead to the development of a projects observatories model to guide the construction of these observatories could be carried out.

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